

QUADRATIC FUNCTION PROJECT

1. Find a parabola in a work of art or architecture. Take or find a picture that gives a good frontal view of the parabola. (Here is a website you may want to take a look at: <http://compfight.com/search/parabola/1-2-1-1>.)
2. Draw a coordinate system over the picture (you may need to enlarge the photo; here is a free photo editor <http://www.gimp.org/downloads/>).
 - a. Mark the x and y axes.
 - b. Mark the scale.
3. Find three points on the graph that can be used to find the equation of the parabola in the form $f(x) = a(x - h)^2 + k$.
4. Use three different points on the graph to check your equation (they only need to be approximations).
5. Identify the following:
 - a. The vertex.
 - b. All intercepts.
 - c. The maximum and minimum.
 - d. The domain and the range.
6. Write the equation in the form $f(x) = ax^2 + bx + c$. Show all your work.
7. Write an essay or prepare a presentation.
 - a. Describe how you completed each of the above steps.
 - b. Show all your work from the above steps.
 - c. Include the picture with the coordinate system.